

Metric Mind

From alert to answer — automated root-cause
analysis on ClickHouse

Team-Disha · Click-a-thon 2026 · InMobi track · Ashiq Abdulkhader · Poushali Chattopadhyay

The alert is easy. The “why” is not.

- At InMobi-scale, revenue / fill / eCPM move across thousands of app × device × geo × format slices
- Today: humans drill dashboards for hours (or days)
- Data already exists in the warehouse — the bottleneck is investigation, not collection
- A wrong “why” costs more than a missed alert

Build a system that detects → localizes → explains — with every claim backed by a computed number.

ClickHouse computes. The LLM only narrates.

1. Detect

Same-weekday (T-7) WoW +
seasonality gate + optional
ML residual.

2. Localize

Factor shares (requests / fill /
eCPM) → segment and combo
contribution.

3. Explain

Plain-language diagnosis + rule-
outs + counterfactual “what-
if” revenue.

LibreChat (ask) · Langfuse (prove) · ClickStack (observe)

Stack

LAYER	CHOICE	ROLE
Warehouse	ClickHouse Cloud	Primary store + analytical engine (eda.rca_*)
Orchestration	Python / uv	Materialize, MCP, thin enrichment
Chat demo	LibreChat	Multi-agent RCA (Orchestrator + specialists)
Product traces	Langfuse Cloud	Tools, SQL spans, final answer
Ops telemetry	ClickStack (HyperDX + OTel)	Latency / errors into Cloud otel
LLM	Azure OpenAI (gpt-5.6-sol)	Narration + agent planning only

How the system fits together



- Judges ask in LibreChat → RCA MCP / ClickHouse MCP → eda.rca_*
- clickathon materialize builds anomalies entirely in ClickHouse SQL
- Langfuse = investigation receipt · HyperDX = runtime health

One command builds the investigation

ad_events → metrics_hourly → rca_daily_wow (+ seasonality)



rca_factor_day · rca_segment_day · rca_combo_day



rca_day_signals → rca_incidents → rca_counterfactual



RCA MCP / LibreChat → diagnosis (+ charts)

```
uv run clickathon materialize --rollup
```

How we decide “what broke”

Baseline — same weekday - 7 (glossary-aligned, not trailing average)

Identity — $\text{Revenue} \approx \text{Requests} \times \text{Fill} \times \text{eCPM} / 1000$

Factor shares — which lever moved the money

Localization — OS / region / format / category + combos via dictGet

Seasonality gate — z-score vs prior like-weeks (thin history → allow; rich → suppress noise)

Counterfactual — revenue if fill / eCPM / requests stayed at T-7

ML additive — `simpleLinearRegression(T-7 → actual) + residual_z` in SQL

Numbers never originate in the LLM

ClickHouse SQL → findings JSON → LLM narrator → diagnosis

(+ Langfuse stores JSON + spans)

- Incident catalog = rca_incidents only (agents cannot invent anomalies)
- Quotes must appear in tool output
- Judges re-check with: `uv run python stack/scripts/verify_unseen_rca.py`

Three integrations, three jobs

JOB

LibreChat	Conversational RCA; Orchestrator → Detector / Factor / Localizer
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Langfuse	Product trace: tools + ClickHouse SQL + answer
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ClickStack	Ops: OTel → HyperDX (latency, errors)
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What the system found (unseen · Jul 6–10)

ID	WINDOW	PROBE VS T-7	PRIMARY FACTOR	SEGMENT	SHAPE
A	Jul 6–7	Jul 7 vs Jun 30	eCPM	video × APAC	layered
B	Jul 8–9	Jul 8 vs Jul 1	fill_rate	iOS 17.5	localized
C	Jul 10	Jul 10 vs Jul 3	eCPM	video × APAC	layered

B — fill: iOS 17.5 fill 0.48 vs 0.79 (−30.8 pp); revenue −8.3%. If fill@T-7 → 551 vs 503 (Δ −48).

A — price: Global eCPM −22%, revenue −1.5% (requests +5.6%, volume OK). If eCPM@T-7 → 602 vs 538 (Δ −64).

C — price again: Global eCPM −32%, revenue −7.1%; video × APAC eCPM 2.94 vs 6.99. If eCPM@T-7 → 571 vs 487 (Δ −84).

Full write-up + SQL + public Langfuse traces → [unseen_incident/](#)

Team-Disha

Repo: Team-Disha/ · Run + verify: README
(verify_unseen_rca.py) · Architecture:
Architecture.md · Unseen pack: unseen_incident/

Demo video (primary) · Langfuse public trace · Live
LibreChat: [metric-](#)
[mind.ashiqabulkhader.dev/login](#)